

Course Code:B23IT4101					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
INTERNET OF THINGS					
For IT					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	List any two roles of Wireless Sensor Networks (WSN) in IoT systems.	1	2	2
	b).	State the importance of M2M communication in IoT applications.	1	2	2
	c).	List important sensor characteristics that affect data accuracy in IoT systems.	2	2	2
	d).	State the need for edge computing in IoT environments.	2	2	2
	e).	Define Zigbee and mention one application area of Zigbee in IoT.	3	2	2
	f).	Discuss the need of LoRa in IoT communication.	3	2	2
	g).	Discuss IoT interoperability.	4	2	2
	h).	State the role of fog computing in reducing latency in IoT systems	4	2	2
	i).	State how IoT improves efficiency in smart agriculture systems.	5	2	2
	j).	List any two features of Vehicular IoT (V-IoT).	5	2	2
5 x 10 = 50 Marks					
UNIT-1					
2.	a).	Explain Machine-to-Machine (M2M) communication and its role in IoT.	1	2	5
	b).	Describe the evolution of IoT.	1	2	5
OR					
3.	a).	Explain IoT architecture and networking components.	1	2	5
	b).	Explain addressing strategies in IoT.	1	2	5
UNIT-2					
4.	a).	Explain sensors in IoT and classify them based on power requirements, output, and measured properties.	2	2	5
	b).	Discuss sensorial deviations in IoT systems.	2	2	5
OR					
5.	a).	Explain the different processing topologies in IoT.	2	2	5
	b).	Explain processing offloading in IoT. Describe different offload locations and the factors affecting offloading decisions.	2	2	5

		UNIT-3			
6.	a).	Explain IEEE 802.15.4 and Zigbee.	3	2	5
	b).	Describe LoRa, Sigfox, NB-IoT.	3	2	5
		OR			
7.	a).	Discuss the IoT Data protocols MQTT and CoAP.	3	2	5
	b).	Explain IoT device management & semantic protocols.	3	2	5
		UNIT-4			
8.	a).	Explain IoT interoperability and discuss the challenges that create the need for interoperability in IoT.	4	2	5
	b).	Describe any two standards and any two frameworks used in IoT.	4	2	5
		OR			
9.	a).	Discuss the role of fog computing in IoT.	4	2	5
	b).	Explain applications of fog computing.	4	2	5
		UNIT-5			
10	a).	Explain emerging IoT paradigms.	5	2	5
.	b).	Discuss the challenges associated with IoT.	5	2	5
		OR			
11	a).	Explain future trends in IoT.	5	2	5
.	b).	Describe Agricultural IoT.	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23HS4101					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
HUMAN RESOURCE & PROJECT MANAGEMENT					
(Common to CSE, AIML, IT, AIDS, CSG, CSIT)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Write the role of HR manager.	1	1	2
	b).	Define recruitment.	1	1	2
	c).	List any two models of HR accounting.	2	1	2
	d).	Define career counseling.	2	1	2
	e).	Define Resource management.	3	1	2
	f).	State the 80-20 rule.	3	1	2
	g).	Define Brainstorming.	4	1	2
	h).	Write about Cash flow analysis.	4	1	2
	i).	List any two forms of project organization.	5	1	2
	j).	Define project control.	5	1	2
5 x 10 = 50 Marks					
UNIT-1					
2.	a).	Interpret the nature and scope of Human Resource Management.	1	2	5
	b).	Discuss the emerging trends in HRM and their impact on organizations.	1	2	5
OR					
3.	a).	Explain the ethical aspects of HRM with suitable examples.	1	2	5
	b).	Describe the steps involved in the selection procedure.	1	2	5
UNIT-2					
4.	a).	Discuss the different methods of training used in organizations.	2	2	5
	b).	Explain the importance of performance appraisal in organizations.	2	2	5
OR					
5.	a).	Summarize the concept of career development and its importance.	2	2	5
	b).	Explain group interaction and its importance in HRD.	2	2	5
UNIT-3					
6.	a).	Explain the concept of Project Management and its importance.	3	2	5
	b).	Discuss the different types of projects with examples.	3	2	5
OR					

7.	a).	Explain the stages of the project life cycle.	3	2	5
	b).	Interpret the process of project selection.	3	2	5
		UNIT-4			
8.	a).	Describe the criteria used for project selection.	4	2	5
	b).	A company invests ₹2,00,000 in a project. The expected cash inflows are: Year 1: ₹50,000 Year 2: ₹60,000 Year 3: ₹70,000 Year 4: ₹80,000 Calculate the Payback Period.	4	2	5
		OR			
9.		A project requires ₹2,00,000 investment. It generates ₹80,000 annually for 4 years. Discount rate = 10% Calculate the NPV and state whether the project is acceptable.	4	2	10
		UNIT-5			
10.	a).	Interpret the process of project planning in detail.	5	2	5
	b).	Explain the human aspects of project management.	5	2	5
		OR			
11.	a).	Discuss the prerequisites for successful project implementation.	5	2	5
	b).	Summarize the importance of project review.	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23CS4102					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
SOFTWARE ARCHITECTURE & DESIGN PATTERNS					
(Common to CSE,AIDS, CSBS, CSD)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Explain the role of Model–View–Controller (MVC) in design patterns.	1	2	2
	b).	Differentiate between cohesion and coupling.	1	2	2
	c).	Explain how actors are identified in use-case analysis.	2	2	2
	d).	Explain the concept of conceptual classes in system analysis.	2	2	2
	e).	Summarize the key idea behind the Adapter pattern.	3	2	2
	f).	Describe how the Proxy pattern controls access to another object.	3	2	2
	g).	Explain the role of the Model component in MVC architecture.	4	2	2
	h).	Explain how the undo operation can be supported in an interactive system.	4	2	2
	i).	Explain the concept of a distributed object system.	5	2	2
	j).	Explain the roles of client and server in a distributed application.	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Describe the essential elements used to document a design pattern.	1	2	5
	b).	Describe how design patterns help solve common design problems in software development.	1	2	5
		OR			
3.	a).	Describe the key concepts of object-oriented design.	1	2	5
	b).	Describe the drawbacks of the object-oriented paradigm.	1	2	5
		UNIT-2			
4.	a).	Use requirement gathering techniques to prepare a functional requirement specification for a student attendance management system.	2	3	5
	b).	Demonstrate how use-case analysis can be used to represent the functional requirements of a bank ATM system.	2	3	5
		OR			
5.	a).	Illustrate the process of identifying conceptual classes and relationships from a given problem description.	2	3	5
	b).	Apply domain knowledge to refine the analysis model of an e-commerce order processing system.	2	3	5
		UNIT-3			

6.	a).	Demonstrate how the Bridge pattern can be used to separate device abstraction from its implementation in a remote-control system.	3	3	5
	b).	Illustrate the use of the Decorator pattern to dynamically add additional features to a text editor component.	3	3	5
		OR			
7.	a).	Apply the Facade pattern to simplify the interaction between multiple subsystems in a banking application.	3	3	5
	b).	Demonstrate how the Flyweight pattern can be used to optimize memory usage in a large document editor with many repeated characters.	3	3	5
		UNIT-4			
8.	a).	Apply the MVC architectural pattern to design an interactive drawing application, identifying the model, view, and controller components.	4	3	5
	b).	Apply MVC principles to design the subsystems of an interactive online shopping interface.	4	3	5
		OR			
9.	a).	Demonstrate how a new feature can be added to an MVC-based system without modifying the entire architecture.	4	3	5
	b).	Illustrate how pattern-based solutions improve extensibility in an MVC-based interactive system.	4		5
		UNIT-5			
10.	a).	Demonstrate how Java Remote Method Invocation (RMI) can be used to enable communication between distributed objects in a networked application.	5	3	5
	b).	Demonstrate how RESTful web services can be used to provide data access in a student information system.	5	3	5
		OR			
11.	a).	Illustrate the process of implementing an object-oriented system on the web for an online shopping application.	5	3	5
	b).	Illustrate how an Enterprise Service Bus (ESB) can be used to integrate different enterprise applications in a distributed environment.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

IV B.Tech. I Semester MODEL QUESTION PAPER

DEEP LEARNING

For IT

Time: 3 Hrs.

Max.Marks: 70 M

Answer Question No.1 compulsorily

Answer **ONE** Question from **EACH UNIT**

Assume suitable data if necessary

10 x 2 = 20 Marks

			CO	KL	M
1.	a).	What is probabilistic modelling?	1	1	2
	b).	What is gradient boosting?	1	1	2
	c).	Define ReLU activation function.	2	1	2
	d).	Define Cross-Entropy loss function.	2	1	2
	e).	Define CNTK? Mention its purpose	3	2	2
	f).	What is a convolutional layer?	3	1	2
	g).	Define LSTM.	4	1	2
	h).	What is a sequence-to-sequence model?	4	1	2
	i).	Define Natural Language Processing?	5	2	2
	j).	Define Autoencoder.	5	1	2

5 x 10 = 50 Marks

		UNIT-1			
2.	a).	Describe Random Forest algorithm.	1	2	5
	b).	Explain human learning vs machine learning.	1	2	5
		OR			
3.	a).	Explain probabilistic modelling in machine learning.	1	2	5
	b).	Explain kernel methods with examples.	1	2	5
		UNIT-2			
4.		Discuss dropout, batch normalization, and regularization.	2	2	10
		OR			
5.	a).	Describe different activation functions.	2	2	5
	b).	Describe backpropagation in detail.	2	2	5
		UNIT-3			
6.	a).	Apply Keras to build a simple neural network model.	3	3	5

	b).	Demonstrate multichannel convolution with an example.	3	3	5
		OR			
7.	a).	Apply the steps to configure a deep learning workstation for training a neural network model.	3	3	5
	b).	Apply stride and padding concepts to compute output dimensions of a convolution layer.	3	3	5
		UNIT-4			
8.	a).	Demonstrate sequence-to-sequence model working.	4	3	5
	b).	Demonstrate recursive neural network with example.	4	3	5
		OR			
9.	a).	Demonstrate LSTM architecture with calculations.	4	3	5
	b).	Apply Bidirectional RNN to a sequence problem.	4	3	5
		UNIT-5			
10.	a).	Explain Generative Adversarial Networks (GAN).	5	2	5
	b).	Explain Restricted Boltzmann Machines.	5	2	5
		OR			
11.	a).	Discuss deep reinforcement learning.	5	2	5
	b).	Describe applications of autoencoders.	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code:B23IT4104					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
COMPUTER VISION					
(Common to IT, CSD)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define radiometry and its importance in image formation.	1	1	2
	b).	What is meant by the local shading model? Give an example.	1	2	2
	c).	Define sampling and aliasing.	2	1	2
	d).	What is texture representation in images?	2	1	2
	e).	What is binocular fusion?	3	1	2
	f).	Define image segmentation.	3	1	2
	g).	What is the Hough Transform used for?	4	1	2
	h).	What is the purpose of the EM algorithm?	4	1	2
	i).	Define camera calibration.	5	1	2
	j).	What is radial distortion in cameras?	5	1	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Compare different light sources and their effects on image formation and shadows.	1	3	5
	b).	Explain the working of a pinhole camera model and its limitations in real-world imaging.	1	2	5
		OR			
3.	a).	Describe photometric stereo and its role in recovering surface orientation	1	2	5
	b).	Explain local shading models, to interpret shading variations in an image.	1	2	5
		UNIT-2			
4.	a).	Explain Fourier Transform concepts in analyzing spatial frequency of images.	2	2	5
	b).	Discuss various edge detection techniques and their sensitivity to noise.	2	2	5
		OR			

5.	a).	Explain texture representation methods in texture analysis or synthesis.	2	2	5
	b).	Analyze the effects of sampling and aliasing in image processing with suitable examples.	2	3	5
		UNIT-3			
6.	a).	Explain stereopsis and how depth is reconstructed from two views.	3	2	5
	b).	Analyze the role of human visual perception (Gestalt principles) in grouping and segmentation.	3	3	5
		OR			
7.	a).	Compare graph-theoretic clustering methods for image segmentation.	3	3	5
	b).	Discuss the concept of image segmentation and clustering techniques for segmenting images.	3	2	5
		UNIT-4			
8.	a).	Explain data association in tracking and it in real-world scenarios.	4	2	5
	b).	Discuss model fitting as a probabilistic inference problem and its robustness.	4	2	5
		OR			
9.	a).	Analyze the working of Kalman filtering in tracking dynamic systems.	4	3	5
	b).	Explain the EM algorithm and it to solve missing data problems in segmentation.	4	2	5
		UNIT-5			
10.	a).	Discuss camera calibration techniques and least-squares estimation for parameter calculation.	5	2	5
	b).	Explain perspective projection and its role in camera modeling.	5	2	5
		OR			
11.	a).	Explain radial distortion and its methods to correct it.	5	2	5
	b).	Analyze the role of geometric camera models in applications like mobile robot localization or medical image registration.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code:B23IT4104					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
BLOCKCHAIN TECHNOLOGY & APPLICATIONS					
(Common to IT, AIDS, CSBS)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define Blockchain and list its components.	1	1	2
	b).	What is decentralization in blockchain?	1	2	2
	c).	Define Smart Contract.	2	1	2
	d).	What are Oracles in blockchain?	2	2	2
	e).	What is a Private Blockchain?	3	1	2
	f).	Define Byzantine Fault Tolerance.	3	2	2
	g).	What is Blockchain scalability issue?	4	2	2
	h).	List any two applications of blockchain	4	2	2
	i).	What is Hyperledger Fabric?	5	1	2
	j).	List any two blockchain case study domains.	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Explain the origin and evolution of blockchain technology	1	2	5
	b).	Describe different types of blockchain and consensus mechanisms	1	2	5
		OR			
3.	a).	Explain the structure of a block in a blockchain with a neat diagram	1	2	5
	b).	Discuss cryptocurrency basics and types with suitable examples.	1	2	5
		UNIT-2			
4.		Explain Public Blockchain systems and the working of Bitcoin and Ethereum blockchain	2	3	10
		OR			
5.		Explain Smart Contracts, their characteristics, types, and role of oracles in blockchain	2	3	10
		UNIT-3			

6.	a).	Explain the architecture and characteristics of Private Blockchain systems	3	2	5
	b).	Describe Multichain and Byzantine Fault Tolerance in permissioned blockchain	3	3	5
		OR			
7.	a).	Explain Consortium Blockchain and its need in industry	3	2	5
	b).	Discuss Hyperledger, Ripple, and Corda platforms	3	2	5
		UNIT-4			
8.		Explain security and privacy challenges in blockchain along with identity management and smart contract security	4	3	10
		OR			
9.		Discuss various applications of blockchain in different domains and explain its limitations	4	2	10
		UNIT-5			
10.	a).	Explain blockchain case studies in retail and banking sectors	5	3	5
	b).	Describe blockchain implementation using Python and relevant packages	5	2	5
		OR			
11.	a).	Explain components of Hyperledger Fabric architecture.	5	2	5
	b).	Discuss chaincode and blockchain application development using Fabric Java SDK	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23CS4107					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
AGILE METHODOLOGIES					
(Common to CSE, AIDS, CSBS, IT)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a)	What is meant by Agile manifesto	1	1	2
	b)	Define task board	1	1	2
	c)	Define self organizing team	2	1	2
	d)	Suggest appropriate agile communication principles for a software development team that is struggling with communication gaps and delayed deliveries in Ebook Reader project.	2	3	2
	e)	Mention the three main roles on a Scrum project.	3	1	2
	f)	How is Scrum both incremental and iterative?	3	2	2
	g)	Define pair programming	4	1	2
	h)	What is feedback loops and why do XP teams create them	4	2	2
	i)	Define Shuhari	5	1	2
	j)	What are the types of waste that create constraints in the workflow?	5	1	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a)	Explain the core values of Agile manifesto and describe how they differ from traditional software development approaches	1	2	5
	b)	Analyze how a “fractured perspective” leads to project failure. How does Agile thinking help resolve this issue?	1	3	5
		OR			
3.	a)	Explain the idea of “Principles over Practices” in Agile. Why is focusing only on practices without understanding underlying principles often ineffective	1	2	5
	b)	Illustrate with an example how “Responding to Change over Following a Plan” benefits a software project.	1	3	5
		UNIT-2			
4.		Explain Agile principles and their importance in software development.	2	2	10
		OR			
5.	a)	Explain how effective communication contributes to better project	2	2	5

		delivery in software development.			
	b)	Explain how continuous improvement contributes to better project and team performance in Agile	2	2	5
		UNIT-3			
6.	a)	Explain the Scrum framework. Discuss the roles of Product Owner, Scrum master, and development team	3	2	5
	b)	Design a complete sprint planning process for a mobile application development project. Apply user stories, story points. Velocity estimation and task board management to create a comprehensive sprint plan.	3	3	5
		OR			
7.	a)	Explain the significance of Scrum values. Explain why Scrum practices alone are insufficient without adherence to these values.	3	2	5
	b)	A sprint fails because the Product Owner frequently changes priorities during sprint. Use Scrum planning rules to analyze the product owner support for sprint success without disrupting execution.	3	3	5
		UNIT-4			
8.		Explain the XP values and their role in shaping an effective development mindset	4	2	10
		OR			
9.	a)	Illustrate how continuous integration helps in maintaining software quality in XP	4	2	5
	b)	Explain how XP practices help teams embrace changing requirements effectively.	4	2	5
		UNIT-5			
10.	a)	Explain Lean thinking and identify different types of waste in software development	5	2	5
	b)	Analyse how limiting Work in Progress (WIP) improves flow in a Kanban system	5	3	5
		OR			
11.	a)	Explain Kanban principles to manage workflow in a software maintenance project	5	2	5
	b)	Evaluate the role of Agile Coach in helping organizations adopt agile practices successfully	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

IV B.Tech. I Semester MODEL QUESTION PAPER

BIG DATA ANALYTICS

For IT

Time: 3 Hrs.

Max. Marks: 70 M

Answer Question No.1 compulsorily

Answer **ONE** Question from **EACH UNIT**

Assume suitable data if necessary

10 x 2 = 20 Marks

			CO	KL	M
1.	a).	List the applications of Big Data.	1	2	2
	b).	Discuss the challenges of adopting Big Data Analytics.	1	2	2
	c).	Distinguish any three differences between the NoSQL and relational databases .	2	2	2
	d).	Define Sharding.	2	2	2
	e).	List the steps in the map reduce algorithm.	3	2	2
	f).	Interpret Rack awareness in HDFS	3	2	2
	g).	Describe Hive database.	4	2	2
	h).	Differentiate between spark context and spark session.	4	2	2
	i).	Define lazy evaluation in spark	5	2	2
	j).	Define a Tumbling window.	5	2	2

5 x 10 = 50 Marks

UNIT-1					
1.	a).	Define Unstructured Data and provide three examples. Why is it harder to analyze than structured data?	1	2	5
	b).	How can predictive analytics in healthcare reduce patient readmission rates?	1	2	5
OR					
2.	a).	Discuss how the simultaneous growth of Cloud Computing, Social Media, and the Internet of Things (IoT) led to the "Big Data" era.	1	2	5
	b).	Define Hadoop? Explain the core components of Hadoop.	1	2	5
UNIT-2					
3.	a).	Explain the concept of an aggregate and why it is the basic unit of data transfer in many NoSQL systems.	2	2	5
	b).	Briefly explain the purpose of a materialized view and how it differs from a standard database view.	2	2	5
OR					
4.	a).	Describe how Cassandra uses a Partition Key to determine the placement of data across a cluster.	2	3	5

	b).	Explain the relationship between Sharding and Replication. Why is it common practice to implement both in a single distributed system?	2	2	5
		UNIT-3			
5.	a).	Illustrate the anatomy of a classic Map Reduce Job Run?	3	2	5
	b).	Implement Matrix multiplication using Map Reduce.	3	3	5
		OR			
6.	a).	Discuss the HDFS architecture with a neat diagram.	3	2	5
	b).	Develop Map Reduce code for counting occurrences of specific words in the input text.	3	3	5
		UNIT-4			
7.	a).	Discuss the architecture of Hive along with its features.	4	2	5
	b).	Explain the roles of the Resource Manager, Node Manager, and Application Master when a Spark application is submitted in cluster mode	4	3	5
		OR			
8.	a).	Explain Spark components in detail. Also list the features of spark.	4	2	5
	b).	Explain the concept of Resilient Distributed Dataset (RDD) in Apache Spark. Write a Spark program to demonstrate basic RDD operations.	4	3	5
		UNIT-5			
9.	a).	Explain how serialization is performed for spark applications?	5	2	5
	b).	Explain the concept of stateful processing.	5	2	5
		OR			
10.	a).	Define Watermark and what is its role in the spark environment?	5	2	5
	b).	What are the input sources for structured streaming? Explain.	5	2	5

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M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23IT4107					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
MOBILE COMPUTING					
For IT					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Differentiate mobile computing, ubiquitous computing and pervasive computing.	1	2	2
	b).	Discuss the limitations of handheld devices	1	2	2
	c).	Expand CDMA and mention its use.	2	1	2
	d).	What is GPRS?	2	1	2
	e).	What are the advantages of I-TCP?	3	2	2
	f).	Design goals of Mobile IP.	3	1	2
	g).	Define MANET and mention any two properties of MANET.	3	2	2
	h).	What is synchronization in mobile computing systems?	4	1	2
	i).	Define mobile agents and give one example.	4	2	2
	j).	What is Wireless Datagram Protocol (WDP) in WAP architecture?	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Explain guided and unguided transmission in mobile communication systems.	1	2	5
	b).	Discuss smart mobile systems, handheld devices, and their applications.	1	3	5
		OR			
3.	a).	Discuss signal propagation frequencies, antennae, and modulation techniques used in mobile communication.	1	3	5
	b).	Explain mobile computing architecture and mobile system networks with neat diagram.	1	2	5
		UNIT-2			
4.		Explain GSM system architecture and radio interfaces with neat diagram.	2	2	10
		OR			
5.	a).	Discuss localization and call handling procedures in GSM networks.	2	3	5
	b).	Compare 3G, 4G, 5G, and 6G wireless communication technologies.	2	3	5

		UNIT-3			
6.	a).	Describe packet delivery and handover management in Mobile IP.	3	3	5
	b).	Discuss MANET architecture, properties, applications, and security issues in Ad-hoc networks	3	3	5
		OR			
7.	a).	Explain the concept of Mobile TCP and its importance in mobile communication.	3	2	5
	b).	Explain Agent Discovery and Location Management in Mobile IP.	3	2	5
		UNIT-4			
8.	a).	Explain the need for synchronization in mobile computing systems and its	4	2	5
	b).	Explain mobile agent design and the concept of aglets in detail.	4	3	5
		OR			
9.	a).	Describe usage models for synchronization in mobile applications.	4	3	5
	b).	Discuss conflict resolution strategies used in data synchronization with suitable examples.	4	3	5
		UNIT-5			
10.	a).	Explain the architecture of Wireless LAN (WLAN) in detail, including components like Access Point, Stations, and Basic Service Set (BSS).	5	2	5
	b).	Discuss the role of WTLS, Wireless Transaction Layer, and Wireless Session Layer in WAP.	5	2	5
		OR			
11.	a).	Describe the IEEE 802.11 protocol layers and their functions.	5	2	5
	b).	Explain the WAP 1.1 architecture and its components in detail.	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code:B23IT4108					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
CYBER PHYSICAL SYSTEMS					
(Common to IT, CSE, CSD)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define Cyber Physical system	1	1	2
	b).	What are the Basic Techniques used in Cyber physical systems.	1	1	2
	c).	Define attack models in CPS security	2	1	2
	d).	Justify security importance in cyber-physical systems?	2	2	2
	e).	What is synchronization in distributed CPS?	3	1	2
	f).	Discuss fault tolerance challenges in distributed CPS	3	2	2
	g).	Differentiate between hard real-time and soft real-time systems	4	2	2
	h).	Explain the role of deadlines in scheduling	4	2	2
	i).	What is model integration in CPS?	5	1	2
	j).	Discuss multidisciplinary interactions in CPS design	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Differentiate between traditional control systems and cyber-physical systems	1	2	5
	b).	Explain the motivation behind symbolic control design	1	2	5
		OR			
3.	a).	What are the challenges in controller synthesis for CPS?	1	1	5
	b).	Discuss safety and reachability specifications in CPS	1	2	5
		UNIT-2			
4.	a)	Explain the security challenges of Cyber-Physical Systems.	2	2	5
	b)	What are the major threats to CPS? Explain	2	1	5
		OR			
5.	a).	Explain redundancy as a countermeasure in CPS	2	2	5
	b).	Differentiate between insider and outsider attacks	2	2	5
		UNIT-3			
6.	a).	Discuss reliability and consistency issues in distributed CPS	3	2	5

	b).	Explain event-based synchronization methods	3	1	5
		OR			
7.	a).	Explain the role of formal software engineering in CPS	3	2	5
	b).	Explain the working of consensus algorithms	3	2	5
		UNIT-4			
8.	a).	Define Real-Time Systems and explain their importance in CPS	4	2	5
	b).	Explain fixed timing parameters in real-time scheduling	4	2	5
		OR			
9.	a).	Explain the role of memory in real-time scheduling	4	2	5
	b).	Explain variability and uncertainty in CPS scheduling	4	2	5
		UNIT-5			
10.	a).	Explain the motivation behind integrated modeling approaches	5	2	5
	b).	Discuss components and connectors in CyPhyML	5	2	5
		OR			
11.	a).	Discuss synchronous and asynchronous timing models.	5	2	5
	b).	Explain causal relationships in CPS models	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

